

OPS-RB-002 - Service Recovery Procedure

Operational Procedure | Operations

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Department	Operations	Classification	INTERNAL
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1. Purpose

This procedure defines how Operations restores correct service and correct **data** after a platform failure. It covers the period after the immediate technical cause has been addressed by Engineering and before the incident can be closed.

The distinction it enforces is the reason it exists: a service returning to health is not the same as the data it produced being correct. Most of the recovery work on this platform is data recovery, not service restart.

2. Scope

Applies to any incident where a platform service produced incorrect, stale, duplicated or missing data, or where a business process was interrupted.

Applies to feed pipelines, event consumers, settlement computation, fulfilment integrations and catalogue publication.

Does not cover deployment rollback, which is OPS-RB-005, or escalation routing, which is OPS-RB-003.

3. Preconditions

1. An incident is open with an identified Engineering owner.
2. The immediate cause has been mitigated, or a mitigation plan is agreed.
3. The affected data domain and time window are known, at least

approximately.

4. An Operations recovery owner has been named.

Do not begin data recovery while the cause is still active. Recovering data into a system that is still producing bad data wastes the effort and obscures the boundary of the problem.

4. Responsibilities

Role	Responsibility
Operations recovery owner	Owns the recovery plan, execution and verification sign-off
Engineering incident owner	Confirms the cause is mitigated; supports replay and reprocessing
Support	Owns seller communication throughout recovery
Business	Decides commercial remediation where sellers bear loss
IT	Acts where recovery requires credential or access changes

5. Procedure

Step 1 - Establish the Blast Radius

Before recovering anything, determine the boundary.

1. Identify the **time window**: first and last affected record.
2. Identify the **data domain**: which entities are wrong, and in what way.
3. Identify the **affected accounts**, and whether the pattern is

platform-wide or account-specific.

4. Quantify. A count of affected records is required, not an estimate.

Where the start of the window cannot be established, record that explicitly and use the earliest evidenced point. An undetermined start is itself a finding and belongs in the incident report.

Step 2 - Classify the Data Fault

Fault type	Recovery approach
Stale data	Resynchronise from source after correcting the pipeline
Duplicated data	Identify duplicates by a business key, reverse the surplus
Missing data	Replay from the source or the event log
Incorrect transformation	Correct the logic, then republish the affected population
Unprocessed backlog	Process the backlog, then verify no gap remains

Classification determines the approach and the verification method. Do not begin execution until the fault type is agreed with the Engineering owner.

Step 3 - Decide Whether to Halt the Flow

Ask whether continuing to operate causes further harm.

Halt where continued operation creates irreversible harm: accepting orders against stock known to be wrong, dispatching against an incorrect catalogue, or paying out from a ledger known to be incorrect.

Do not halt where the harm is reporting lag or internal inconvenience and the flow can be corrected in place.

In May 2026 a seller disabled affected listings roughly twenty minutes after stale-stock oversells were confirmed. That decision stopped further oversells for the twenty-four hours needed to correct the pipeline. Halting was correct because an oversell cannot be unwound: the customer has paid for something that does not exist.

Step 4 - Build the Recovery Plan

The plan must state:

1. The recovery approach per Step 2.
2. The order of operations, with dependencies.
3. Whether a replay authorisation is required, and from whom.
4. The verification method and the acceptance threshold.
5. The rollback position if recovery makes matters worse.
6. The seller communication points.

Record the plan before executing it. A recovery executed from memory cannot be verified.

Step 5 - Execute

1. Correct the source of the fault first. Confirm with Engineering.
2. Where reprocessing is required, obtain explicit replay authorisation. A batch carrying a processing token must not be rerun under the same token without a documented reason.
3. Execute in the smallest safe increment. Verify the increment before proceeding.
4. Preserve the incorrect data until verification is complete. Do not delete evidence during recovery.

Step 6 - Verify

Verification is against the **source of truth**, not against the system that failed.

Fault type	Verification
Stale data	Published position compared against the seller's own export
Duplicated data	Count of records per business key equals one
Missing data	Record count reconciled against source or event log
Incorrect transformation	Population-level comparison against source distribution
Unprocessed backlog	Consumer lag at zero and no gap in the sequence

For transformation faults, per-record sampling is not sufficient. A transformation defect can leave every individual record plausible while shifting the whole population; compare distributions.

Step 7 - Remediate Affected Parties

1. Identify customers and sellers who bore a loss.
2. Refer commercial remediation to Business. Operations does not decide who absorbs a cost.
3. Where recovery requires money to be reclaimed from a seller, stage it across settlement cycles unless the seller requests otherwise. A single large debit on a cash-constrained account causes real harm for no benefit.
4. Where an amount is uneconomic to pursue, recommend write-off with a rationale.

Step 8 - Close

1. Confirm verification passed and record the evidence.
2. Confirm preventive actions are raised with owners and dates.
3. Confirm the seller has been told the outcome, not only the cause.
4. Sign off recovery and hand the incident back to the Engineering owner for report completion.

6. Decision Points

Decision	Criterion	Owner
Halt the flow	Continued operation causes irreversible harm	Operations, with seller where applicable

Decision	Criterion	Owner
Authorise a replay	Cause confirmed mitigated and a documented reason exists	Operations
Accept verification	Comparison against source within the stated threshold	Operations
Write off an amount	Recovery cost exceeds the amount, or pursuit is disproportionate	Business
Close recovery	All verification evidence recorded	Operations

7. Exceptions

Recovery may be closed with a known residual where the residual is quantified, accepted by a named owner, and recorded in the incident report. An unquantified residual is not an acceptable exception.

Recovery may be deferred where executing it during a peak period carries more risk than the incorrect data itself. This requires Business agreement, because the deferral is a commercial choice.

8. Escalation

Situation	Escalate to	Timeframe
Blast radius cannot be established	Engineering incident owner	Immediately
Recovery would require halting a revenue flow	Business	Before halting
Verification fails	Engineering incident owner, halt recovery	Immediately
Recovery requires reclaiming money from a seller	Business	Before communicating to the seller
Recovery exceeds its planned window by more than 50 percent	Operations lead	On breach

9. Post-Incident Actions

- 1. Record the recovery method and its verification in the incident report.
 - 2. Identify whether the fault was detectable. If detection was by a seller, raise a detection gap as a preventive action.
 - 3. Identify whether a silent fallback or a permissive default hid the fault.
- These are the most common contributing factor in this platform's incidents.
- 4. Confirm the requirement that was breached exists as an enforced control, not only as a written requirement.

10. Revision History

Version	Date	Change
1.0	2024-02-14	Initial procedure
2.0	2025-07-09	Separated data recovery from service restoration

Version	Date	Change
2.1	2026-06-08	Replay authorisation and processing tokens added
2.2	2026-08-28	Population-level verification required for transformation faults
2.3	2026-09-18	Staged recovery of amounts reclaimed from sellers